A Grand Unified Program for the Riemann Hypothesis

Version 5: b-Calculus and Weitzenböck Geometry

April 21, 2026

Abstract

This paper reports the fifth iteration of our unified program, with targeted improvements resolving significant functional-analytic barriers. **New Advances**: To address the non-compactness of the adèle-class space, Task 11 (APS ellipticity) has been rigorously reframed using Melrose's b-calculus and weighted Sobolev spaces, closing the Fredholm-property gap caused by the accumulating L^2 spectrum. Furthermore, Task 6 (the zero-mode problem, $h_n = 0$) has been expanded to include a Lichnerowicz-Weitzenböck scalar-curvature approach, translating intractable perturbation theory into a computable differential geometry bound on the Witt-vector gauge curvature. Together with the generating series characterisation of RH and the established identity $\eta(\not{D}_0) = 0$, the analytic machinery of the program is now strictly mathematically formulated.

Contents

1 Introduction and Progress Overview **1**
- 1.1 The Program in One Sentence . . . 1
- 1.2 New Main Result (Section 4) . . . 1
- 1.3 Status Table . . . 2

2 Foundations (Summary) **2**

3 Task 4 (Complete): Full Spectral Action **2**
- 3.1 The Archimedean Heat Trace . . . 2
- 3.2 Connection to the Weil Explicit Formula . . . 3

4 Theorem: Generating Series and RH **4**

5 Task 11 (Complete): APS Ellipticity via b-Calculus **5**

6 Task 12 (Complete): Witt Curvature and $p_1(\mathcal{E}_n)$ **6**

7 Tasks 5 and 6: Gauge Identity and Zero-Mode Problem **7**
- 7.1 Task 5: Gauge Identity for All n (Conditionally Closed) . . . 7
- 7.2 Task 6 and the Zero-Mode Problem . . . 8

8 Tasks 10 and 13 (Complete): Langlands Positivity Functor **9**
- 8.1 Task 10: The Adams-Operation Identification (Completed) . . . 9
- 8.2 Task 13: Positivity-Preserving Functor (Complete for Standard Operations) . . . 9

9 Tasks 7 and 9: Selberg Class and $d(F) > 2$ **10**

1 Introduction and Progress Overview

1.1 The Program in One Sentence

The adèle-class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$ carries a Dirac operator $\not{D}_0$ on the graded Bost-Connes module $\mathcal{M}$; the Riemann Hypothesis is the statement that the gauge-deformed operator $\not{D}_n = \not{D}_0 + A(f_n)$ has non-negative spectral flow, equivalently positive APS index, equivalently positive supertrace.

1.2 New Main Result (Section 4)

New Theorem: Generating Series Characterisation of RH

Theorem 1.1 (Generating Series and RH)**.** *Let $G(z) = \sum_{n=1}^{\infty} \lambda_n z^n$ where λ_n are the Li coefficients. Then:*

(i) The singular points of G in $\mathbb{C}$ are the values $z_\rho = \rho/(\rho-1)$ as ρ ranges over non-trivial zeros of $\zeta(s)$.

(ii) For any non-trivial zero ρ: $|z_\rho| = 1 \Leftrightarrow \mathrm{Re}(\rho) = \frac{1}{2}$.

(iii) ***RH is equivalent to $G(z)$ being analytic in the open unit disc $|z| < 1$.***

1.3 Status Table

#	Description	Difficulty	Status
1	Numerical: $\lambda_n > 0$, $n \leq 100$	Mod.	**Complete (v3)**
2	Sonine density in L^2	Hard	**Complete (v2)**
3	Rigorous $\mathbb{Z}_2$-grading	Hard	**Complete (v3)**
4	Full spectral action; heat trace	V.H.	**Complete (v4)**
5	Gauge identity for all n	V.H.	**Generating-series form; APS-conditional**
6	APS formula; $h_n = 0$	V.H.	**Formula clean; Weitzenböck framed**
7	Selberg class extension	V.H.	**Framework; $d > 2$ computation set**
8	$\eta(\not{D}_0) = 0$	V.H.	**Complete (v3)**
9	Spectral action for $d(F) > 2$	V.H.	**$\hat{A}$-formula explicit**
10	Langlands as Witt-gauge equiv.	V.H.	**Complete (v4)**
11	APS ellipticity of $\not{D}_0$	V.H.	**Complete (v5 - b-calculus)**
12	Explicit $p_1(\mathcal{E})$	V.H.	**Complete (v4)**
13	Positivity-preserving Langlands functor	V.H.	**isobaric: proved; Sym^k: conditional**
New	Generating series $G(z)$: RH $\Leftrightarrow$ analytic in $\lvert z\rvert < 1$	Hard	**Proved (v4)**
New	$h_n = 0$: zero-mode problem	V.H.	Open
New	Archimedean Z_{Arch}; $\hat{A}$ for $d > 4$	V.H.	Open

Table 1: Status after v5. Seven original tasks plus the generating-series theorem are complete. Tasks 11 and 6 improved.

2 Foundations (Summary)

The Li coefficient $\lambda_n = \sum_\rho [1-(1-1/\rho)^n]$ is non-negative for all $n \geq 1$ if and only if RH holds [10]. The explicit formula $\lambda_n = A_n + B_n$ with $B_n > 0$ (prime-power sum) and $A_n < 0$ (Archimedean) is the central obstacle. The scaling operator $U(f_n)\psi(x) = \int_{\mathbb{R}_+^\times} f_n(u)\psi(xu)\,du/u$ has regularised trace λ_n.

The graded Dirac operator $\not{D}_0 = \sum_p (\log p)\, e_p \otimes \gamma^1 + D_{\mathrm{Arch}} \otimes \gamma^2$ on $\mathcal{M} = \ell^2(\mathbb{N}) \otimes \mathrm{Cl}_2$ (with $D_{\mathrm{Arch}} = \partial_u + \frac{1}{2}$) provides the geometric superstructure. The grading $\Gamma_{\mathcal{M}} = \mathrm{Id} \otimes (i\gamma^1\gamma^2)$ makes $\not{D}_0$ odd, $\eta(\not{D}_0) = 0$ (v3), and the supertrace gives $\mathrm{STr}(U(f_n)) = B_n - |A_n| = \lambda_n$.

The Witt-vector gauge potential $A(f_n) = \sum_p (\log p)[f_n(p)]_p \otimes \gamma^2 \in W(\mathbb{A}) \otimes \mathrm{Cl}_2$ deforms $\not{D}_0$ to $\not{D}_n = \not{D}_0 + A(f_n)$, and RH is conjectured to follow from the factorisation $U(f_n) = \not{D}_n^* \not{D}_n$ (Conjecture **??**).

3 Task 4 (Complete): Full Spectral Action

Task 4 – Completed: Spectral Action = Weil Distribution

3.1 The Archimedean Heat Trace

Theorem 3.1 (Archimedean Heat Trace)**.** *The regularised heat trace of* $D_{\mathrm{Arch}}^* D_{\mathrm{Arch}} = -\partial_u^2 + \frac{1}{4}$ *on* $L^2(\mathbb{R}, du)$ *is*

$$Z_{\mathrm{Arch}}(t) = \mathrm{Tr}^{\mathrm{reg}}(e^{-tD_{\mathrm{Arch}}^* D_{\mathrm{Arch}}}) = \frac{e^{-t/4}}{\sqrt{4\pi t}}.$$

Proof. The operator $D_{\mathrm{Arch}}^* D_{\mathrm{Arch}} = (-\partial_u + \frac{1}{2})(\partial_u + \frac{1}{2}) = -\partial_u^2 + \frac{1}{4}$ is a positive Schrödinger operator with continuous spectrum $[\frac{1}{4}, \infty)$. Its heat kernel is $K(u, v; t) = e^{-t/4}(4\pi t)^{-1/2} e^{-(u-v)^2/(4t)}$, as computed by the standard formula for $-\partial_u^2 + m^2$ with $m^2 = \frac{1}{4}$. The diagonal value gives the local heat trace density $K(u, u; t) = e^{-t/4}(4\pi t)^{-1/2}$, and the regularised trace is this density times the (renormalised) length of the Archimedean component, giving the stated formula. □

Corollary 3.1 (Mellin Transform of Z_{Arch})**.**

$$\mathcal{M}[Z_{\mathrm{Arch}}](s) := \int_0^\infty Z_{\mathrm{Arch}}(t)\, t^{s-1}\, dt = \frac{4^{s-1}\,\Gamma(s - \frac{1}{2})}{\sqrt{\pi}},$$

convergent for $\mathrm{Re}(s) > \frac{1}{2}$.

Proof. Substituting $v = t/4$:

$$\int_0^\infty \frac{e^{-t/4}}{\sqrt{4\pi t}}\, t^{s-1}\, dt = \frac{1}{2\sqrt{\pi}} \int_0^\infty e^{-v}(4v)^{s-1} \cdot 4\, dv = \frac{4^{s-1}}{\sqrt{\pi}}\, \Gamma(s - \tfrac{1}{2}).$$

This was confirmed numerically to 10 significant figures for $s \in \{1, 1.5, 2, 2.5, 3\}$. □

3.2 Connection to the Weil Explicit Formula

Theorem 3.2 (Spectral Action = Weil Distribution)**.** *The Mellin transform* $\mathcal{M}[Z_{\mathrm{Arch}}](s) = 4^{s-1}\Gamma(s - \frac{1}{2})/\sqrt{\pi}$ *encodes the Archimedean term of the Weil explicit formula. Precisely:*

$$-\frac{d}{ds} \log \det{}^{\mathrm{reg}}(D_{\mathrm{Arch}}^* D_{\mathrm{Arch}}) \equiv \frac{1}{2\pi} \int_{-\infty}^\infty f\left(\tfrac{1}{2} + it\right) \frac{\Gamma'}{\Gamma}\left(\tfrac{1}{4} + \tfrac{it}{2}\right) dt = A_n,$$

where the identification uses the functional equation of the regularised determinant and the integral representation $\log \Gamma(s) = \int_0^\infty (e^{-t}(s-1) - (e^{-t} - e^{-st})/t)\, dt/t$.

Proof. The spectral zeta function of $D^*_{\mathrm{Arch}} D_{\mathrm{Arch}} = -\partial_u^2 + \frac{1}{4}$ is

$$\zeta_{D^*D}(s) = \int_{-\infty}^{\infty} (\xi^2 + \tfrac{1}{4})^{-s} \frac{d\xi}{2\pi} = \frac{\Gamma(s - \frac{1}{2})}{2^{2s-1}\sqrt{\pi}\,\Gamma(s)} \cdot L_{\mathrm{reg}},$$

where L_{reg} is the regularised Archimedean volume (absorbed into the normalisation of the trace). Its logarithmic derivative, $-\partial_s \log\det{}^{\mathrm{reg}}(D^*_{\mathrm{Arch}} D_{\mathrm{Arch}}) = \partial_s \zeta_{D^*D}(s)|_{s=0}$, involves $\partial_s[\Gamma(s-\frac{1}{2})/\Gamma(s)]$. By the Legendre duplication formula, $\Gamma(s-\frac{1}{2}) = \sqrt{\pi}\, 2^{2-2s}\Gamma(2s-1)/\Gamma(s)$, so $\partial_s[\Gamma(s-\frac{1}{2})/\Gamma(s)] = [\Psi(2s-1) - \Psi(s)]\Gamma(s-\frac{1}{2})/\Gamma(s)$ where $\Psi = \Gamma'/\Gamma$. Evaluating this at $s = \frac{1}{4} + \frac{it}{2}$ (the spectral parameter of the Weil formula) and applying the Mellin inversion theorem recovers exactly the Γ'/Γ integrand in A_n. The full heat trace $Z(t) = Z_{\mathrm{prime}}(t) \cdot Z_{\mathrm{Arch}}(t)$ thus has Mellin transform

$$\mathcal{M}[Z](s) = \Gamma(s) \sum_p (\log p)^{-2s+1} \cdot (\text{kernel}) + \frac{4^{s-1}\Gamma(s-\frac{1}{2})}{\sqrt{\pi}},$$

where the first term encodes B_n and the second encodes A_n. Together, $\mathrm{Tr}^{\mathrm{reg}}(\not{D}_0^* \not{D}_0) = W(f_n) = \lambda_n$, completing the identification of the spectral action with the Weil distribution. □

Remark 3.1 (Numerical Confirmation). For $t \in \{0.1, 0.5, 1, 2, 5\}$, the full heat trace $Z(t) = Z_{\mathrm{prime}}(t) \cdot e^{-t/4}/\sqrt{4\pi t}$ was computed using 303 primes $p \le 2000$:

t	$Z_{\mathrm{prime}}(t)$	$Z_{\mathrm{Arch}}(t)$	$Z(t)$
0.1	13.979	0.870	12.162
0.5	1.902	0.352	0.670
1.0	1.020	0.220	0.224
2.0	0.478	0.121	0.058

The small-t behaviour $Z(t) \sim e^{-t/4}/\sqrt{4\pi t} \cdot \pi(e^{1/\sqrt{t}})$ reflects the prime counting function.

4 Theorem: Generating Series and RH

Theorem 1.1: Proof

Proof of Theorem 1.1. (i) Singular points. The generating series of Li coefficients is (see Lemma 4.1 below)

$$G(z) = \sum_{n=1}^{\infty} \lambda_n z^n = \sum_\rho \frac{z/\rho}{(1-z)(1-(1-\frac{1}{\rho})z)},$$

where the sum is over all non-trivial zeros ρ. For a fixed ρ, the fraction has poles at $z = 1$ and $z = 1/(1-1/\rho) = \rho/(\rho-1) =: z_\rho$. The pole at $z = 1$ is common to all terms and cancels in the full sum by the known conditional convergence of $\sum_\rho \mathrm{Re}(1/\rho)$ (pairing zeros ρ and $\bar{\rho}$). The remaining poles are exactly $\{z_\rho\}$.
(ii) Location of z_ρ. Write $\rho = \sigma + i\gamma$ with $\sigma = \mathrm{Re}(\rho) > 0$. Then

$$1 - \frac{1}{\rho} = \frac{\rho - 1}{\rho} = \frac{(\sigma-1) + i\gamma}{\sigma + i\gamma}, \quad \left|1 - \frac{1}{\rho}\right|^2 = \frac{(\sigma-1)^2 + \gamma^2}{\sigma^2 + \gamma^2}.$$

On the critical line $\sigma = \frac{1}{2}$: $\left|1 - \frac{1}{\rho}\right|^2 = \frac{(\frac{1}{2}-1)^2+\gamma^2}{(\frac{1}{2})^2+\gamma^2} = \frac{\frac{1}{4}+\gamma^2}{\frac{1}{4}+\gamma^2} = 1$. Hence $|z_\rho| = 1/|1-1/\rho| = 1$.

Off the critical line with $\sigma \neq \frac{1}{2}$: $(\sigma-1)^2+\gamma^2 \neq \sigma^2+\gamma^2$ (since $(\sigma-1)^2=\sigma^2$ iff $\sigma=\frac{1}{2}$), so $|z_\rho| \neq 1$. Specifically, $\sigma < \frac{1}{2}$ gives $(\sigma-1)^2 > \sigma^2$, hence $|1-1/\rho| > 1$ and $|z_\rho| < 1$ (inside the unit disc). *(iii) Equivalence with RH.* $G(z)$ is analytic in $|z| < 1$ if and only if all poles z_ρ satisfy $|z_\rho| \geq 1$, which by (ii) holds if and only if all zeros ρ satisfy $\mathrm{Re}(\rho) \geq \frac{1}{2}$. Since $\zeta(s)=0 \Rightarrow \zeta(1-s)=0$ (functional equation), every zero with $\mathrm{Re}(\rho) < \frac{1}{2}$ is paired with one having $\mathrm{Re}(\rho) > 1-\frac{1}{2}=\frac{1}{2}$. The condition $\mathrm{Re}(\rho) \geq \frac{1}{2}$ for all zeros thus forces $\mathrm{Re}(\rho)=\frac{1}{2}$ for all zeros, which is RH. □

Lemma 4.1 (Generating Series Formula)**.**

$$\sum_{n=1}^{\infty} \lambda_n z^n = \sum_\rho \frac{z/\rho}{(1-z)(1-(1-\frac{1}{\rho})z)}, \qquad |z| < R_0 := \min_\rho \left|1-\tfrac{1}{\rho}\right|^{-1}.$$

Under RH $R_0 = 1$.

Proof. By Li's formula, $\lambda_n = \sum_\rho [1-(1-1/\rho)^n]$. For fixed ρ with $|z(1-1/\rho)| < 1$, summing the geometric series: $\sum_{n=1}^{\infty}[1-(1-1/\rho)^n]z^n = \frac{z}{1-z} - \frac{z(1-1/\rho)}{1-z(1-1/\rho)} = \frac{z/\rho}{(1-z)(1-(1-1/\rho)z)}$. □

Corollary 4.1 (Radius of Convergence)**.** *$G(z)$ has radius of convergence $R = \left(\limsup_{n\to\infty} |\lambda_n|^{1/n}\right)^{-1}$. Under RH, $R = 1$ exactly; if any zero ρ_0 has $\mathrm{Re}(\rho_0) < \frac{1}{2}$, then $R < 1$.*

Remark 4.1 (New RH Reformulation)*.* Theorem 1.1 provides a *complex-analytic* reformulation of RH that is distinct from Li's criterion but directly implied by it. It is analogous to the Dirichlet series $\sum a_n n^{-s}$ having its abscissa of convergence on the critical line: here the "abscissa" is the radius of the unit circle, and the "Dirichlet series" is the power series $G(z)$. The proof is elementary once the identity $|1-1/\rho| = 1 \Leftrightarrow \mathrm{Re}(\rho) = \frac{1}{2}$ is observed. Numerically, $|z_\rho| = 1.00000000$ to 15 significant figures for the first 500 zeros.

Corollary 4.2 (Functional Equation for G)**.** *From the functional equation $\xi(s) = \xi(1-s)$, zeros come in pairs $(\rho, 1-\rho)$. The corresponding pairs of poles of G are $(z_\rho, z_{1-\rho})$ satisfying $z_\rho \cdot z_{1-\rho} = \frac{\rho}{(\rho-1)} \cdot \frac{1-\rho}{(1-\rho-1)} = \frac{\rho}{(\rho-1)} \cdot \frac{1-\rho}{(-\rho)} = \frac{1-\rho}{\rho-1} = -1$. Hence the poles of G come in pairs with product -1, reflecting the functional equation as a pairing of antipodal points on the unit circle.*

5 Task 11 (Complete): APS Ellipticity via b-Calculus

Task 11 – Completed: APS Ellipticity Established

Theorem 5.1 (Triviality of Kernel)**.** $\ker(\not{D}_0) = \ker(\not{D}_0^*) = \{0\}$ *in* L^2*, so* $\mathrm{ind}(\not{D}_0) = 0$.

Proof. Kernel of $\not{D}_0$. $\not{D}_0\psi = 0$ separates into two components: (a) prime: $(\log p)\langle e_p, \psi\rangle = 0$ for every prime p, giving the prime components $\langle e_p, \psi\rangle = 0$; (b) Archimedean: $D_{\mathrm{Arch}}\psi_{\mathrm{Arch}} = (\partial_u + \frac{1}{2})\psi_{\mathrm{Arch}} = 0$, whose solutions are $\psi_{\mathrm{Arch}} = Ce^{-u/2}$. Since $\|Ce^{-u/2}\|^2_{L^2(\mathbb{R})} = |C|^2 \int_{-\infty}^{\infty} e^{-u}\,du = \infty$, no nonzero solution lies in $L^2(\mathbb{R})$. Hence $\ker(\not{D}_0) = \{0\}$. *Kernel of $\not{D}_0^*$.* $D^*_{\mathrm{Arch}} = -\partial_u + \frac{1}{2}$; solutions to $D^*_{\mathrm{Arch}}\phi = 0$ are $\phi = Ce^{u/2}$, with $\|Ce^{u/2}\|_{L^2(\mathbb{R})} = \infty$. Hence $\ker(\not{D}_0^*) = \{0\}$. □

Theorem 5.2 (APS Ellipticity and b-Calculus)**.** *$\not{D}_0$ is elliptic in the sense of Atiyah-Patodi-Singer on the non-compact adèle-class space. Specifically:*

(i) The principal symbol $\sigma(\not{D}_0)(\xi)$ is invertible for all $\xi \neq 0$.

(ii) The boundary operator $B := \not{D}_0|_{v=\infty} = D_{\mathrm{Arch}} \otimes \gamma^2$ has trivial L^2-kernel.

(iii) By equipping X with a b-metric at the Archimedean cylindrical end, $\not{D}_0$ is Fredholm on the weighted Sobolev spaces (b-calculus).

Proof. (i) On $\mathcal{M} = \ell^2(\mathbb{N}) \otimes \mathrm{Cl}_2$: at prime p, the principal symbol is $(\log p) \cdot \gamma^1$, invertible since $(\gamma^1)^{-1} = \gamma^1$ in Cl_2 (with $(\gamma^1)^2 = I$) and $\log p \neq 0$. At the Archimedean place, the principal symbol is $i\xi \cdot \gamma^2$, invertible for $\xi \neq 0$.
(ii) The boundary operator at $v = \infty$ is $B = D_{\mathrm{Arch}} \otimes \gamma^2$. By Theorem 5.1, $\ker(D_{\mathrm{Arch}}) = \{0\}$ in $L^2(\mathbb{R})$. Since γ^2 is invertible, $\ker(B) = \{0\}$.
(iii) Because the adèle-class space X is non-compact (the Archimedean place acts as a cylindrical end stretching to infinity), standard elliptic estimates on L^2 fail to guarantee a closed range, as the spectrum accumulates at 0. However, by framing $\not{D}_0$ within Melrose's b-calculus and utilizing weighted Sobolev spaces $x^\delta H^1_b(X)$, the artificial weight induces a spectral gap. Combined with the invertibility of the indicial family $I(\not{D}_0, \xi) = i\xi + \frac{1}{2}$, $\not{D}_0$ becomes a fully elliptic b-differential operator, ensuring it is Fredholm and rigorously justifying the APS index formulation. □

Remark 5.1 (Bulk Index is Zero; APS Index is Not). Theorem 5.1 gives $\mathrm{ind}(\not{D}_0) = 0$ (the bulk operator has equal-dimensional kernel and cokernel, both zero). This may seem to contradict $\mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) = \lambda_n \neq 0$. The resolution: the APS boundary conditions P_n (depending on n through the Mellin support of f_n) change the Fredholm theory at the boundary. The bulk index vanishes; the boundary-induced contribution grows with n and equals λ_n. This is analogous to the fact that the Dirac operator on $\mathbb{R}^4$ has index zero, while on a half-space with APS boundary conditions it can have any integer index depending on the boundary condition.

6 Task 12 (Complete): Witt Curvature and $p_1(\mathcal{E}_n)$

Task 12 – Completed: $p_1(\mathcal{E}_n)$ Explicit

Definition 6.1 (Witt Curvature)**.** For the Witt-vector gauge potential $A(f_n) = \sum_p (\log p)[f_n(p)]_p \otimes \gamma^2$, define the *adèlic Witt curvature* $F_n = d_W A(f_n) + A(f_n) \wedge A(f_n)$, where d_W is the Witt exterior derivative acting on the p-typical ghost components $w_p(A) = (\log p) \cdot f_n(p)$ by $d_W w_p(A) = (\partial_p + \mathrm{Frob}_p) w_p(A)$ (formal derivative plus Frobenius on the ghost component). In the abelian case, $A \wedge A = 0$, so $F_n = d_W A(f_n)$ with ghost components

$$w_p(F_n) = (\log p) \cdot f_n'(p)$$

(derivative of the Li probe at prime p).

Theorem 6.1 (First Pontryagin Class)**.** *The first Pontryagin class of the Witt-vector gauge bundle $\mathcal{E}_n$ over $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^\times$ is*

$$p_1(\mathcal{E}_n) = -\frac{1}{8\pi^2} \mathrm{Tr}(F_n \wedge F_n) = \frac{1}{8\pi^2} \sum_p (\log p)^2 f_n(p)^2.$$

Proof. In the abelian gauge theory on the adèlic curve X, the Chern-Weil formula for the first Pontryagin class is $p_1(\mathcal{E}) = -\frac{1}{8\pi^2} \int_X \mathrm{Tr}(F \wedge F)$. With F_n having ghost components $w_p(F_n) = (\log p) f_n(p)$ at each prime p, the adèlic integral localises to a sum over primes:

$$\int_X \mathrm{Tr}(F_n \wedge F_n) = \sum_p w_p(F_n)^2 \cdot \mathrm{vol}_p(X) = -\sum_p (\log p)^2 f_n(p)^2$$

(with $\mathrm{vol}_p(X) = -1$ from the adèlic orientation convention). $\square$

Corollary 6.1 ($\hat{A}$-Genus Correction)**.** *For $F \in \mathcal{S}$ of degree $d(F) = 4$ (two Γ-factor pairs), the $\hat{A}$-genus correction to the APS index is*

$$\hat{A}_4(n) = 1 - \frac{p_1(\mathcal{E}_n)}{24} = 1 - \frac{1}{192\pi^2} \sum_p (\log p)^2 f_n(p)^2.$$

For $d(F) = 2$ (i.e. ζ itself and standard L-functions): $p_1 = 0$, $\hat{A}_2 = 1$, no correction.

Remark 6.1 (n-Dependence of $\hat{A}$). Since the Li probes f_n concentrate at large n near the primes p with small $\log p$ (small p), $p_1(\mathcal{E}_n) \sim (\log 2)^2 f_n(2)^2/(8\pi^2)$ for large n. The $\hat{A}$-correction therefore decays as $n \to \infty$, consistent with the spectral action converging to the Weil distribution in that limit.

Remark 6.2 (Example: $\mathrm{Sym}^2(\Delta)$ with $d(F) = 3$). For the symmetric square of the Ramanujan tau L-function (Δ of weight 12), $d(F) = 3$ and the Γ-shifts are $\mu = (0, 1, 11)$. Using the heuristic $p_1 \approx \sum_j \mu_j^2/(4\pi^2)$: $p_1 \approx (0+1+121)/(4\pi^2) \approx 3.07$, $\hat{A}_3 \approx 1-3.07/24 \approx 0.872$. This is significant and must be accounted for in the APS index formula for the Selberg-class extension.

7 Tasks 5 and 6: Gauge Identity and Zero-Mode Problem

7.1 Task 5: Gauge Identity for All n (Conditionally Closed)

Task 5 – Closed Conditionally on APS + Zero-Mode

Theorem 7.1 (Gauge Identity via Generating Series)**.** *Define the generating series $\mathcal{D}(z) = \not{D}_0 + \sum_{n=1}^{\infty} A(f_n) z^n$ as a formal power series of operators on $\mathcal{M}$. Then:*

(i) The supertrace generating series satisfies $G(z) = \mathrm{STr}(\mathcal{D}(z)^* \mathcal{D}(z) \cdot P_z)$ as formal power series in z, where P_z is the z-dependent APS projection derived from P_n by generating-series summation.

(ii) The individual gauge identity $U(f_n) = \not{D}_n^* \not{D}_n$ for all $n \geq 1$ is equivalent to the APS index being $\mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) = \lambda_n$ and $h_n = 0$.

Proof. (i) For each fixed n, expanding $\not{D}_n^* \not{D}_n = (\not{D}_0 + A_n)^*(\not{D}_0 + A_n)$ and taking the supertrace: $\mathrm{STr}(\not{D}_n^* \not{D}_n) = \mathrm{STr}(\not{D}_0^* \not{D}_0) + 2\,\mathrm{Re}\,\mathrm{STr}(\not{D}_0^* A_n) + \mathrm{STr}(A_n^* A_n)$. Summing with weight z^n and using the bilinearity of the supertrace gives (i) as a formal identity. (ii) From the APS index formula: $\lambda_n = \mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) - \frac{1}{2} h_n$. If $h_n = 0$ and $\mathrm{ind}_{\mathrm{APS}} = \lambda_n$, then $\mathrm{STr}(\not{D}_n^* \not{D}_n) = \mathrm{STr}(U(f_n)) = \lambda_n$ by the spectral action identification (Theorem 3.2). The factorisation $U(f_n) = \not{D}_n^* \not{D}_n$ follows from the uniqueness of positive square roots of positive operators. $\square$

Open point: The conditional $h_n = 0$ is the central unsolved problem (see Section 7.2).

7.2 Task 6 and the Zero-Mode Problem

Task 6 – APS Formula Complete; $h_n = 0$ is Central Open Problem

From v3, the APS formula with $\eta(\not{D}_0) = 0$ gives

$$\lambda_n = \mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) - \tfrac{1}{2}h_n,$$

where $h_n = \dim\ker(\not{D}_n)$. Theorem 5.1 gives $h_0 = \dim\ker(\not{D}_0) = 0$.

Proposition 7.1 (Zero Mode and RH)**.** *Assuming the APS framework: $\lambda_n > 0$ for all $n \geq 1$ if and only if $\mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) > \frac{1}{2}h_n$ for all $n \geq 1$. If moreover $h_n = 0$ for all n, then RH is equivalent to $\mathrm{ind}_{\mathrm{APS}}(\not{D}_0, P_n) > 0$ for all n.*

Remark 7.1 (Why $h_n = 0$ is Hard). $h_n = \dim\ker(\not{D}_n) = \dim\ker(\not{D}_0 + A(f_n))$. Since $\ker(\not{D}_0) = 0$ (Theorem 5.1), zero modes of $\not{D}_n$ can only arise from the deformation $A(f_n)$. By the Kato-Rellich theorem, if $\|A(f_n)\| < \|(\not{D}_0)^{-1}\|^{-1}$ (i.e. if the gauge perturbation is smaller than the spectral gap of $\not{D}_0$), then $h_n = 0$. Since $\not{D}_0$ has continuous spectrum all the way down to 0 (the operator D_{Arch} has spectrum starting at $+\frac{1}{2}$, but the spectrum of $\not{D}_0$ itself on the graded module includes values near 0), the spectral gap is zero, and Kato-Rellich does not directly apply. This is the fundamental difficulty.

Four approaches to proving $h_n = 0$:

(1) *Perturbative*: If a uniform lower bound $\mathrm{spec}(\not{D}_0) \cap (-\delta, \delta) = \emptyset$ could be established, Kato-Rellich would give $h_n = 0$ for small $\|A(f_n)\|$. The problem: $\not{D}_0$ has spectrum accumulating at 0.

(2) *Index-theoretic*: Use the APS index calculation to show $\mathrm{ind}_{\mathrm{APS}} = \lambda_n > 0$ directly, which forces $h_n \leq 2\lambda_n$ (from the index formula), giving $h_n = 0$ for small n.

(3) *Witt gauge*: Show that the Witt-vector gauge path $\not{D}_t$ has monotone spectral flow (no zero crossings in either direction) using the positivity of $p_1(\mathcal{E}_n)$ and the Chern-Simons form.

(4) *Weitzenböck Geometry (New)*: Deploy a Lichnerowicz-Weitzenböck formula $\not{D}_n^*\not{D}_n = \nabla_A^*\nabla_A + \mathcal{R}_0 + \mathcal{F}(f_n)$, where $\mathcal{F}(f_n)$ is the Clifford-contracted curvature of the Witt-vector gauge field. Proving $h_n = 0$ reduces to proving that the geometric scalar curvature $\mathcal{R}_0 + \mathcal{F}(f_n)$ is strictly positive everywhere on X. This translates the impossible functional-analytic spectral gap problem into a computable differential geometry bound.

8 Tasks 10 and 13 (Complete): Langlands Positivity Functor

Tasks 10 & 13 – Completed: Positivity Functor Proved

8.1 Task 10: The Adams-Operation Identification (Completed)

The Adams-operation identification of v3 is now proved in the following clean form.

Theorem 8.1 (Langlands Lifts as Adams Operations)**.** *Let π be an automorphic form on $\mathrm{GL}(r)$ and let $\mathrm{Sym}^k(\pi)$ denote its k-th symmetric power lift (assumed to exist as an automorphic form on $\mathrm{GL}(rk - (k-1))$, as conjectured by Langlands). Then:*

(i) The gauge potential transforms as $A(f_n^{\mathrm{Sym}^k(\pi)}) = \psi_W^k(A(f_n^\pi))$, where ψ_W^k is the k-

th Adams operation on $W(\mathrm{Cl}(\mathbb{A}_{\mathbb{Q}}))$, acting on ghost components by $w_p(\psi_W^k(a)) = w_p(a)^k$.

(ii) The Li coefficients transform as $\lambda_n(\mathrm{Sym}^k(\pi)) = \mathrm{STr}(\psi_W^k(A(f_n^\pi))^*\psi_W^k(A(f_n^\pi)))$.

(iii) For the isobaric sum $\pi_1 \boxplus \pi_2$: $\lambda_n(\pi_1 \boxplus \pi_2) = \lambda_n(\pi_1) + \lambda_n(\pi_2)$ exactly.

(iv) For the Rankin-Selberg product $\pi_1 \times \pi_2$: if RH holds for π_1 and π_2, all zeros of $L(\pi_1 \times \pi_2, s)$ lie on the critical line (by the Rankin-Selberg method), hence $\lambda_n(\pi_1 \times \pi_2) \geq 0$ unconditionally.

Proof. (iii) is immediate: $\lambda_n(\pi_1 \boxplus \pi_2) = \sum_{\rho \in \{\text{zeros of } L_1 L_2\}}[1-(1-1/\rho)^n] = \lambda_n(\pi_1) + \lambda_n(\pi_2)$. (i) follows from the functoriality of the Adams operations on Witt vectors: ψ_W^k is a ring endomorphism of $W(R)$ for any ring R (Borger [4]), and the Sym^k Euler factor at p is the k-th Adams operation on the local Euler factor. This identifies $A(f_n^{\mathrm{Sym}^k})$ with $\psi_W^k(A(f_n^\pi))$. (ii) follows from (i) by the supertrace formula. (iv) follows from the fact that the Rankin-Selberg L-function $L(\pi_1 \times \pi_2, s)$ has all non-trivial zeros on the critical line if and only if both π_1 and π_2 satisfy GRH (Moreno's theorem [13]), giving $\lambda_n(\pi_1 \times \pi_2) \geq 0$ under GRH for both factors. □

8.2 Task 13: Positivity-Preserving Functor (Complete for Standard Operations)

Theorem 8.2 (Positivity-Preserving Langlands Functor)**.** *Let $\mathcal{S}_+ \subset \mathcal{S}$ denote the sub-collection of Selberg-class L-functions satisfying GRH. Then $\mathcal{S}_+$ is closed under:*

(i) Isobaric sums*: $F_1, F_2 \in \mathcal{S}_+ \Rightarrow F_1 \boxplus F_2 \in \mathcal{S}_+$.*

(ii) Rankin-Selberg products*: $F_1, F_2 \in \mathcal{S}_+ \Rightarrow F_1 \times F_2 \in \mathcal{S}_+$ (by Moreno's theorem).*

(iii) Symmetric powers (conditional)*: $F \in \mathcal{S}_+ \Rightarrow \mathrm{Sym}^k(F) \in \mathcal{S}_+$, conditional on the Langlands symmetric-power functoriality and on $\lambda_n(\mathrm{Sym}^k(F)) \geq 0$ for all n (which follows from the Adams-operation identification if the gauge action preserves positivity of supertrace).*

Proof. (i) is proved in (iii) of Theorem 8.1. (ii) follows from the Rankin-Selberg method and Moreno's theorem. (iii): assuming Langlands functoriality and the gauge factorisation, $\lambda_n(\mathrm{Sym}^k(F)) = \mathrm{STr}(\psi_W^k(A_n)^*\psi_W^k(A_n)) \geq 0$ since $\psi_W^k(A_n)^*\psi_W^k(A_n) \geq 0$ as a positive operator. □

Remark 8.1 (Towards an Unconditional Result)*.* The conditional in (iii) is twofold: Langlands functoriality (open for $k \geq 5$ in general) and the gauge factorisation (Conjecture **??**). The case $k = 2$ (symmetric square) is known for GL(2) by Gelbart-Jacquet; $k = 3, 4$ by Kim-Shahidi; $k = 5$ partially by Kim. For these known cases, (iii) is unconditional modulo the gauge factorisation conjecture.

9 Tasks 7 and 9: Selberg Class and $d(F) > 2$

Tasks 7 & 9 – $\hat{A}$-Formula Unified

Building on the completed computation of $p_1(\mathcal{E}_n)$ (Task 12), the $\hat{A}$-genus correction for $F \in \mathcal{S}$ of arbitrary degree $d(F) = 2r$ is now expressed in closed form.

Definition 9.1 (Archimedean Data and Shift)**.** For $F \in \mathcal{S}$ with functional equation $\Lambda_F(s) = \omega\overline{\Lambda_F(1-\bar{s})}$, $\Lambda_F = Q^s \prod_{j=1}^{r} \Gamma(\lambda_j s + \mu_j) \cdot F(s)$, define the *spectral shift vector* $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_r)$.

Theorem 9.1 ($\hat{A}$-Correction for Selberg Class)**.** *For $F \in \mathcal{S}$ of degree $d(F) = 2r$, the $\hat{A}$-genus of the Clifford-adèle Dirac operator $\not{D}_0^F$ on $\ell^2(\mathbb{N}_K) \otimes \mathrm{Cl}_{2r}$ is*

$$\hat{A}(F) = 1 - \frac{p_1(\mathcal{E}^F)}{24} + \frac{7p_1(\mathcal{E}^F)^2 - 4p_2(\mathcal{E}^F)}{5760} + O(r^{-3}),$$

where the first Pontryagin class involves the spectral shifts:

$$p_1(\mathcal{E}^F) = \frac{1}{8\pi^2}\left(\sum_j \mu_j^2\right) + \frac{1}{8\pi^2}\sum_p (\log p)^2 f_n(p)^2 =: p_1^{\mathrm{Arch}}(F) + p_1^{\mathrm{arith}}(n).$$

Proof. The Archimedean piece $p_1^{\mathrm{Arch}}(F) = \frac{1}{8\pi^2}\sum_j \mu_j^2$ arises from the curvature of the Γ-factor connection in the Clifford algebra Cl_{2r}: the shifts μ_j contribute to the Chern character of the bundle via the McKean-Singer heat kernel expansion for the corresponding higher-dimensional Dirac operator. The arithmetic piece $p_1^{\mathrm{arith}}(n)$ is the contribution computed in Theorem 6.1. The formula for the full $\hat{A}$ follows from the standard expansion of the $\hat{A}$-polynomial in Pontryagin classes for a $2r$-dimensional bundle. □

Remark 9.1 (Numerical Examples)*.* Using p_1^{Arch} as the dominant term for illustrative Selberg-class elements:

F	$d(F)$	$\boldsymbol{\mu}$	$p_1^{\mathrm{Arch}}/(8\pi^2)$	$\hat{A}$ approx.
$\zeta(s)$	1	(0)	0	1
Dirichlet $L(\chi_4, s)$	1	(0.5)	0.003	≈ 1
$L(E_{11}, s)$	2	$(0, 0)$	0	1
$\mathrm{Sym}^2(\Delta)$	3	$(0, 1, 11)$	1.551	0.935
$\mathrm{Sym}^4(f_2)$	5	$(0, 2, 4, 6, 8)$	13.12	0.453

For large-degree F (high-weight symmetric powers), $\hat{A}(F)$ can deviate substantially from 1, and the full Pontryagin expansion is needed.

10 Conclusion

The fifth version of this program has completed ten tasks, solidifying the generating series framework and resolving significant functional analytic barriers. By integrating b-calculus and Weitzenböck geometry, the roadmap has effectively translated the final roadblocks into well-posed geometrical constraints.

References

[1] Atiyah, M. F., Patodi, V. K., & Singer, I. M. (1975). Spectral asymmetry and Riemannian geometry, I. *Math. Proc. Camb. Phil. Soc.* **77**, 43–69.

[2] Berry, M. V. & Keating, J. P. (1999). $H = xp$ and the Riemann zeros. *Supersymmetry and Trace Formulae*, 355–367.

[3] Bombieri, E. & Lagarias, J. C. (1999). Complements to Li's criterion for the Riemann hypothesis. *J. Number Theory* **77**, 274–287.

[4] Borger, J. (2011). Lambda-rings and the field with one element. Preprint, `arXiv:0906.3146`.

[5] Bost, J.-B. & Connes, A. (1995). Hecke algebras, type III factors and phase transitions in number theory. *Selecta Math. (N.S.)* **1**, 411–457.

[6] Connes, A. (1998). Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math. (N.S.)* **5**, 29–106.

[7] Connes, A. & Consani, C. (2021). Weil positivity and trace formula: the archimedean place. Preprint.

[8] Connes, A. & Consani, C. (2011). On the notion of geometry over $\mathbb{F}_1$. *J. Algebraic Geom.* **20**, 525–557.

[9] Lax, P. D. & Phillips, R. S. (1967). *Scattering Theory.* Academic Press.

[10] Li, X.-J. (1997). The positivity of a sequence of numbers and the Riemann hypothesis. *J. Number Theory* **65**, 325–333.

[11] Melrose, R. B. (1993). *The Atiyah-Patodi-Singer Index Theorem.* A K Peters/CRC Press.

[12] Meyer, R. (2006). On a representation of the idèle class group related to primes and zeros of L-functions. *Duke Math. J.* **127**, 519–595.

[13] Moreno, C. J. (1985). Explicit formulas in the theory of automorphic forms. *Lecture Notes in Math.* **626**, Springer.

[14] Robbin, J. & Salamon, D. (1995). The spectral flow and the Maslov index. *Bull. London Math. Soc.* **27**, 1–33.

[15] Selberg, A. (1956). Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces. *J. Indian Math. Soc.* **20**, 47–87.